

SymPy Bug Report

1 Bug 31: solveset Returns Real Solutions for an Impossible $\arccos(\cos(u))$ Equation

1.1 Minimal Reproducer

```
from sympy import Eq, S, acos, cos, solveset, symbols

u = symbols("u")
print(solveset(Eq(acos(cos(u)), -1), u, S.Reals))
```

1.2 Observed Behavior

[illegible]

SymPy returns two infinite real periodic families. Representative returned values such as $u = 1$ and $u = -1$ do not satisfy the original equation:

$$\text{acos}(\cos(1)) - (-1) = 2, \quad \text{acos}(\cos(-1)) - (-1) = 2.$$

Thus the returned set contains false positive real solutions.

1.3 Expected Behavior

The correct result is $\emptyset$, represented in SymPy as `EmptySet`. Over the real domain, there is no real u satisfying

$$\text{acos}(\cos(u)) = -1.$$

1.4 Mathematical Explanation

For real u , $\cos(u) \in [-1, 1]$. The principal real arccosine maps this interval into

$$\text{acos}([-1, 1]) = [0, \pi].$$

Consequently,

$$\arccos(\cos(u)) \in [0, \pi]$$

for every real u . The left-hand side is never negative, so it cannot equal -1 . The reported periodic families are not an equivalent branch representation or a harmless constant-shift form: substituting a representative from the returned set into the original equation gives a nonzero residual.

1.5 Source-Level Diagnosis

The diagnosis identifies the relevant logic in `sympy/solvers/solveset.py`, especially the generic inverse-function branch around line 230 and the real trigonometric inversion block around line 410. The traced call path is: the public `solveset` call reaches `_solveset(..., _check=True)`; `_solveset` rewrites the equation as `acos(cos(u)) + 1`; it then calls `_invert`; because the domain is real, `_invert` dispatches to `_invert_real`.

Inside `_invert_real`, the additive constant is stripped, reducing the problem to `acos(cos(u)) = -1`. The generic inverse-function branch then applies `acos.inverse()`, namely `cos`, to the target -1 without first checking whether -1 is in the principal real range of `acos`. This changes the problem into `cos(u) = cos(-1)`, equivalently `cos(u) = cos(1)`. The later real trigonometric inversion code then correctly solves that transformed cosine equation and returns periodic real families. By that point, however, the invalid outer inverse step has already lost the range condition from the original equation.

The diagnosis also notes that the trigonometric range check performed later checks the transformed target $\cos(1) \in [-1, 1]$, which is true; it does not know that the original target -1 was outside the principal range $[0, \pi]$ of `acos`. A plausible fix direction is to avoid handling inverse trigonometric functions through this generic inverse branch unless the right-hand side is first constrained to the inverse function's principal range; for `acos`, the real target set should be intersected with $[0, \pi]$ before applying `cos`.

1.6 Additional Failing Instances

The same failure appears for the family

$$\text{solveset}(\text{acos}(\cos(u)) = -a, u, \mathbb{R}) \quad \text{with} \quad a \in \{1, 2, 3, 4, 5, 6\}.$$

In each case, the right-hand side is negative while $\text{acos}(\cos(u)) \in [0, \pi]$ for all real u , so the expected result is $\emptyset$. SymPy instead returns periodic real families solving the transformed cosine equation.

Input / Expression	SymPy behavior	Expected behavior
$\text{acos}(\cos(u)) = -1$	$\{2\pi n + 1 : n \in \mathbb{Z}\} \cup \{2\pi n - 1 + 2\pi : n \in \mathbb{Z}\}$	$\emptyset$
$\text{acos}(\cos(u)) = -2$	$\{2\pi n + 2 : n \in \mathbb{Z}\} \cup \{2\pi n - 2 + 2\pi : n \in \mathbb{Z}\}$	$\emptyset$
$\text{acos}(\cos(u)) = -3$	$\{2\pi n + 3 : n \in \mathbb{Z}\} \cup \{2\pi n - 3 + 2\pi : n \in \mathbb{Z}\}$	$\emptyset$
$\text{acos}(\cos(u)) = -4$	$\{2\pi n + 4 : n \in \mathbb{Z}\} \cup \{2\pi n - 4 + 2\pi : n \in \mathbb{Z}\}$	$\emptyset$
$\text{acos}(\cos(u)) = -5$	$\{2\pi n + 5 : n \in \mathbb{Z}\} \cup \{2\pi n - 5 + 2\pi : n \in \mathbb{Z}\}$	$\emptyset$
$\text{acos}(\cos(u)) = -6$	$\{2\pi n + 6 : n \in \mathbb{Z}\} \cup \{2\pi n - 6 + 2\pi : n \in \mathbb{Z}\}$	$\emptyset$

1.7 Related Issues Assessment

The bug-finding harness performed a best-effort deduplication check. No exact public SymPy issue, pull request, Stack Overflow report, mailing-list discussion, or release-note entry was identified for `solveset(Eq(acos(cos(u)), -1), u, S.Reals)` or for this periodic `ImageSet` output. The closest related public material concerns the general mathematics of inverse-trigonometric principal ranges. The verdict is therefore a new instance of a known failure mode: the broad inverse-trigonometric range issue is familiar, but this exact reproducible `solveset` case is previously unreported in the available evidence and remains individually actionable as an issue and regression test.

1.8 Proposed SymPy Regression Test

```
import pytest
from sympy import EmptySet, Eq, S, acos, cos, solveset, symbols

@pytest.mark.parametrize("a", [1, 2, 3, 4, 5, 6])
def test_solveset_acos_cos_negative_target_is_empty(a):
    u = symbols("u")
    assert solveset(Eq(acos(cos(u)), -a), u, S.Reals) == EmptySet
```